

An Adaptive Funnel-Based Stopping Strategy with No-Realized-Loss Exits

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Abstract

We propose a threshold-based stopping strategy for trading under drift and volatility. The method compares a log-price path against a drift-adjusted predictive funnel and exits a position only when three conditions are simultaneously satisfied: the price crosses the upper funnel boundary, local momentum weakens, and the log-return exceeds a transaction-cost-adjusted profit threshold. This gives a rigorous no-realized-loss property for completed algorithmic exits. The guarantee does not imply that the strategy is risk-free: mark-to-market wealth may decline while holding, capital may be locked for long periods, and forced exits may realize losses. The framework is formulated using discrete-time stochastic processes, stopping times, and elementary statistical estimation.

1 Introduction

A common trading intuition is to hold a stock while it is below or near the purchase price, and to sell only after a sufficiently favorable move once short-term upward momentum has weakened. The purpose of this paper is to formalize this idea as a rule-based stopping strategy rather than to claim global optimality.

The strategy combines three components:

1. a drift-adjusted deviation statistic measuring how far the log-price has moved relative to its expected path;
2. a local momentum statistic detecting whether recent upward movement has stalled;
3. an explicit profit threshold ensuring that completed algorithmic exits do not realize a loss after transaction costs.

The central point is that the strategy can impose a *no voluntary realized loss* constraint, but it cannot eliminate economic risk. While a position remains open, the mark-to-market value of the position can fall. Thus the strategy transforms realized-loss risk into holding-time risk, opportunity cost, and mark-to-market drawdown.

2 Price Model

Let $(S_t)_{t \geq 0}$ denote the asset price at discrete times. Define the log-price

$$X_t = \log S_t.$$

For the baseline model, assume

$$X_{t+1} = X_t + \mu + \sigma \varepsilon_{t+1}, \quad \varepsilon_{t+1} \sim \mathcal{N}(0, 1),$$

where $\mu \in \mathbb{R}$ is the drift and $\sigma > 0$ is the volatility. Let

$$\mathcal{F}_t = \sigma(X_0, \dots, X_t)$$

be the information available at time t .

3 Deviation Signal and Predictive Funnel

Fix a buying time τ_b . More generally, let μ_s and σ_s denote the drift and volatility estimates used for step s . Define the predictive center and cumulative variance by

$$C(t) = X_{\tau_b} + \sum_{s=\tau_b+1}^t \mu_s, \quad V(t) = \sum_{s=\tau_b+1}^t \sigma_s^2.$$

In the fixed-parameter baseline model, $\mu_s \equiv \mu$ and $\sigma_s \equiv \sigma$, so these reduce to

$$C(t) = X_{\tau_b} + \mu(t - \tau_b), \quad V(t) = \sigma^2(t - \tau_b).$$

For a threshold parameter $k > 0$, define the upper and lower funnel boundaries by

$$U(t) = C(t) + k\sqrt{V(t)}, \quad L(t) = C(t) - k\sqrt{V(t)}.$$

Equivalently, define the normalized deviation statistic

$$Z_t = \frac{X_t - C(t)}{\sqrt{V(t)}}, \quad t > \tau_b.$$

Then $X_t > U(t)$ is equivalent to $Z_t > k$.

Remark 1 (Fixed-time versus sequential interpretation). *For a fixed time t , under the fixed-parameter Gaussian baseline with correctly specified μ and σ , $Z_t \sim \mathcal{N}(0, 1)$. However, the trading rule monitors Z_t repeatedly over time. Therefore the relevant object is not only the fixed-time tail probability $P(Z_t > k)$, but the boundary-crossing probability*

$$P\left(\sup_{\tau_b < t \leq T} Z_t > k\right).$$

Thus k should be interpreted as a funnel-width or boundary-crossing tuning parameter, not simply as a fixed-time significance level.

4 Momentum Signal

For a window length $\delta \geq 1$, define the finite-difference momentum signal

$$M_t = X_t - X_{t-\delta}.$$

When $M_t > 0$, the recent log-price movement is upward over the last δ steps. When $M_t \leq 0$, recent upward momentum has stalled or reversed. Under the baseline Gaussian model,

$$M_t \sim \mathcal{N}(\mu\delta, \sigma^2\delta).$$

Hence M_t is a noisy local trend signal.

5 Transaction Costs and Profit Threshold

Suppose that buying incurs proportional cost $c_{\text{buy}} \geq 0$ and selling incurs proportional cost $c_{\text{sell}} \in [0, 1)$. Buying one unit at price S_{τ_b} effectively costs

$$(1 + c_{\text{buy}})S_{\tau_b},$$

and selling at price S_t yields

$$(1 - c_{\text{sell}})S_t.$$

To avoid a realized loss after transaction costs, it is necessary and sufficient that

$$(1 - c_{\text{sell}})S_t > (1 + c_{\text{buy}})S_{\tau_b}.$$

Taking logarithms gives the profit condition

$$X_t - X_{\tau_b} > h,$$

where

$$h = \log \left(\frac{1 + c_{\text{buy}}}{1 - c_{\text{sell}}} \right).$$

When transaction costs are ignored, $h = 0$.

6 Stopping Rule

Definition 1 (Adaptive funnel stopping rule). *Given a buying time τ_b , define the selling time by*

$$\tau_s = \inf \{t > \tau_b : X_t > U(t), \quad M_t \leq 0, \quad X_t - X_{\tau_b} > h\}.$$

Equivalently,

$$\tau_s = \inf \{t > \tau_b : Z_t > k, \quad M_t \leq 0, \quad X_t - X_{\tau_b} > h\}.$$

The rule says: sell only after the price crosses the upper funnel boundary, local momentum has weakened, and the trade is profitable after costs.

7 Practical Bounded-Funnel and Trailing Exit

The since-entry funnel above is mathematically clean, but in long upward trends its cumulative variance $V(t)$ can grow so large that the upper boundary becomes difficult to cross. In implementation, we therefore use a bounded reference window for the diagnostic funnel. Given a maximum lookback L , define

$$\rho_t = \max\{\tau_b, t - L\}.$$

The displayed and tested funnel may then use

$$C_{\rho_t}(t) = X_{\rho_t} + \sum_{s=\rho_t+1}^t \mu_s, \quad V_{\rho_t}(t) = \sum_{s=\rho_t+1}^t \sigma_s^2,$$

and

$$Z_t^{(L)} = \frac{X_t - C_{\rho_t}(t)}{\sqrt{V_{\rho_t}(t)}}.$$

This keeps the diagnostic funnel from widening indefinitely while preserving the original entry price τ_b for profit accounting.

We also add a trailing-profit exit. Let

$$H_t = \max_{\tau_b \leq u \leq t} X_u$$

be the highest log-price observed since entry, and let $a > 0$ be a trailing drawdown threshold. A voluntary trailing exit is allowed when

$$H_t - X_t \geq a \quad \text{and} \quad X_t - X_{\tau_b} > h.$$

Thus the complete practical sell rule exits when either the bounded-funnel sell condition holds or the trailing-profit condition holds. In both cases, the transaction-cost profit condition $X_t - X_{\tau_b} > h$ is retained, so the no-realized-loss property for completed voluntary exits is unchanged.

8 Trading Cycle and Re-Entry Rule

The stopping rule above describes when an existing long position is exited. To turn the rule into a repeated trading strategy, we also specify how the strategy re-enters after a completed sale.

Suppose the strategy sells at time τ_s and then holds cash. During the cash period, take τ_s as the new reference time, define the corresponding predictive center $C_{\tau_s}(t)$ and variance $V_{\tau_s}(t)$, and set

$$Z_t^{\text{down}} = \frac{X_t - C_{\tau_s}(t)}{\sqrt{V_{\tau_s}(t)}}, \quad t > \tau_s.$$

The original symmetric re-entry rule buys after a lower-funnel recovery:

$$\tau_b^{\text{next}} = \inf \left\{ t > \tau_s : Z_t^{\text{down}} < -k, \quad M_t \geq 0 \right\}.$$

This is useful in mean-reverting or choppy markets, but it can leave the strategy in cash during persistent upward trends. To reduce this opportunity-cost problem, the implementation also allows a trend-continuation re-entry. Let $z_{\text{trend}} \geq 0$ be a smaller entry threshold. The practical re-entry rule is

$$\tau_b^{\text{next}} = \inf \left\{ t > \tau_s : \left(Z_t^{\text{down}} < -k, \quad M_t \geq 0 \right) \text{ or } \left(Z_t^{\text{down}} > z_{\text{trend}}, \quad M_t > 0 \right) \right\}.$$

The first condition is the bargain re-entry rule: buy after an unusually low price has started to recover. The second condition is the trend-continuation rule: buy when the price is above the adaptive center and local momentum is positive. In the simulation, z_{trend} is intentionally smaller than k , because it is meant to detect participation in an ongoing trend rather than an extreme statistical event.

Thus the full algorithm alternates between two regimes:

1. while holding the asset, sell only when the upper-funnel, weakened-momentum, and profit-threshold conditions are all satisfied;
2. while in cash, buy either after a lower-funnel recovery or after a trend-continuation signal.

The no-realized-loss theorem below applies to completed sell exits. The re-entry rule determines when capital is redeployed, but it does not by itself create a realized gain or loss.

9 No-Realized-Loss Property

Theorem 1 (No realized loss on completed algorithmic exits). *Suppose a trade is entered at time τ_b and exited at a finite time $\tau_s < \infty$ according to the stopping rule above. Then the net realized wealth multiplier satisfies*

$$\frac{(1 - c_{\text{sell}})S_{\tau_s}}{(1 + c_{\text{buy}})S_{\tau_b}} > 1.$$

Thus every completed algorithmic exit has positive realized return after transaction costs.

Proof. By the stopping rule,

$$X_{\tau_s} - X_{\tau_b} > h = \log \left(\frac{1 + c_{\text{buy}}}{1 - c_{\text{sell}}} \right).$$

Therefore

$$\frac{S_{\tau_s}}{S_{\tau_b}} = \exp(X_{\tau_s} - X_{\tau_b}) > \frac{1 + c_{\text{buy}}}{1 - c_{\text{sell}}}.$$

Multiplying both sides by $(1 - c_{\text{sell}})/(1 + c_{\text{buy}})$ gives

$$\frac{(1 - c_{\text{sell}})S_{\tau_s}}{(1 + c_{\text{buy}})S_{\tau_b}} > 1.$$

□

Remark 2. *This theorem applies only to completed algorithmic exits satisfying the stopping rule. If a finite horizon T is imposed and the position is forcibly liquidated at T , the theorem need not apply. Forced exits can realize losses.*

10 Wealth Processes

It is important to distinguish realized wealth from mark-to-market wealth. Let W_t^r denote realized wealth, updated only after completed exits. While holding a position purchased at τ_b , define mark-to-market wealth by

$$W_t^m = W_{\tau_b}^r \frac{S_t}{S_{\tau_b}}$$

in the frictionless case. With transaction costs interpreted as liquidation costs, one may instead use

$$W_t^m = W_{\tau_b}^r \frac{(1 - c_{\text{sell}})S_t}{(1 + c_{\text{buy}})S_{\tau_b}}.$$

The no-realized-loss theorem concerns W_t^r , not W_t^m . While holding, W_t^m may decrease.

A natural benchmark is buy-and-hold wealth,

$$W_t^{\text{bh}} = W_0 \frac{S_t}{S_0},$$

which should be compared against both realized and mark-to-market strategy wealth.

11 Momentum Decision Consistency

The finite-difference signal M_t is a noisy local trend statistic. A simple consistency result can be obtained by working with returns. Suppose that over a local window,

$$r_i = X_i - X_{i-1} = \beta + \epsilon_i,$$

where ϵ_i are independent with mean zero and finite variance. Define

$$\hat{\beta}_n = \frac{1}{n} \sum_{i=1}^n r_i.$$

Theorem 2 (Consistency of local mean-return decision). *Under the assumptions above,*

$$\hat{\beta}_n \xrightarrow{p} \beta.$$

Consequently, for any threshold $c \neq \beta$,

$$\mathbf{1}\{\hat{\beta}_n > c\} \xrightarrow{p} \mathbf{1}\{\beta > c\}.$$

Proof. The convergence $\hat{\beta}_n \xrightarrow{p} \beta$ follows from the law of large numbers. Since $c \neq \beta$, the indicator map is continuous at β , and the conclusion follows from the continuous mapping theorem. $\square$

This theorem justifies a smoothed momentum rule based on average recent returns. The simpler finite-difference signal $M_t = X_t - X_{t-\delta}$ corresponds to the cumulative return over the last δ steps and is noisier, but follows the same intuition.

12 Phase 2 Evaluation Metrics

The Phase 2 simulation evaluates the strategy by separating accounting quantities that can otherwise be confused. The baseline simulation tracks three wealth curves:

1. realized wealth W_t^r , which updates only after completed exits;
2. mark-to-market wealth W_t^m , which shows hidden drawdowns while holding;
3. buy-and-hold wealth W_t^{bh} , which measures opportunity cost.

From these curves, the main diagnostic quantities are:

$$R_T^r = \frac{W_T^r}{W_0} - 1, \quad R_T^m = \frac{W_T^m}{W_0} - 1, \quad R_T^{\text{bh}} = \frac{W_T^{\text{bh}}}{W_0} - 1.$$

The benchmark gap is measured by

$$G_T = \frac{W_T^m}{W_T^{\text{bh}}} - 1,$$

so $G_T > 0$ means the strategy's liquidating mark-to-market value exceeds buy-and-hold at the horizon, while $G_T < 0$ means the strategy underperforms buy-and-hold.

To measure hidden risk, define the mark-to-market drawdown process

$$D_t^m = \frac{W_t^m}{\max_{0 \leq s \leq t} W_s^m} - 1,$$

and its maximum drawdown over the horizon

$$D_{\min}^m = \min_{0 \leq t \leq T} D_t^m.$$

The same calculation can be applied to W_t^{bh} to compare the strategy's drawdown against the passive benchmark.

Finally, let $H_t \in \{0, 1\}$ denote whether the strategy is holding the asset. The exposure ratio

$$E_T = \frac{1}{T+1} \sum_{t=0}^T H_t$$

measures the fraction of time capital is exposed to the asset. Together, R_T^r , R_T^m , R_T^{bh} , G_T , $D_{\min}^m$, the number of completed trades, and E_T form the Phase 2 evaluation dashboard.

The simulation should use a Gaussian log-return model when validating the theory. If clipped shocks are used for visualization, they should be labelled as a visualization choice rather than as the exact model assumed in the paper.

13 Phase 3 Funnel Visualization

Phase 3 makes the stopping rule visually explicit. Instead of plotting only the price path and the derived statistic Z_t , the simulation also plots the predictive funnel directly on the price chart.

After a reference time τ , which is either the current buying time while holding or the most recent selling time while in cash, define the log-funnel center and cumulative variance by

$$C_\tau(t) = X_\tau + \sum_{s=\tau+1}^t \mu_s, \quad V_\tau(t) = \sum_{s=\tau+1}^t \sigma_s^2.$$

In the fixed-parameter simulation, where $\mu_s \equiv \mu$ and $\sigma_s \equiv \sigma$, this is equivalent to $C_\tau(t) = X_\tau + \mu(t - \tau)$ and $V_\tau(t) = \sigma^2(t - \tau)$.

The log-funnel boundaries are

$$U_\tau(t) = C_\tau(t) + k\sqrt{V_\tau(t)}, \quad L_\tau(t) = C_\tau(t) - k\sqrt{V_\tau(t)}, \quad t \geq \tau.$$

The corresponding price-level curves are

$$\exp(C_\tau(t)), \quad \exp(U_\tau(t)), \quad \exp(L_\tau(t)).$$

These are the curves displayed in the price panel.

While the strategy is holding the asset, the upper boundary $\exp(U_{\tau_b}(t))$ is the relevant exit boundary. A sell can occur only after the price crosses this upper boundary, momentum has weakened, and the profit-threshold condition is satisfied. While the strategy is in cash, the lower boundary $\exp(L_{\tau_s}(t))$ remains the bargain re-entry boundary. A buy can occur either after the price crosses below this lower boundary and local momentum recovers, or after the price moves above the adaptive center with positive momentum according to the trend-continuation rule.

Whenever the strategy sells or buys, the reference time is reset. Thus the plotted funnel is not one fixed band drawn from $t = 0$; it is a sequence of local predictive funnels, each anchored at the current trade reference point. This visualization is useful because it separates three ideas that are otherwise easy to confuse:

1. the expected log-price path, shown by the funnel center;
2. statistically unusual movement, shown by crossing the upper or lower boundary;
3. executable trading decisions, which also require momentum and, for sells, the profit threshold.

14 Phase 4 Adaptive Drift Estimation

Phase 4 removes the unrealistic assumption that the strategy knows the true drift μ . The simulated market may still be generated from a baseline drift in order to create controlled experiments, but the trading rule uses an estimated drift when drawing its funnel.

Let

$$r_t = X_t - X_{t-1}$$

denote the one-step log-return. We model the drift used by the strategy as a slowly changing latent state:

$$\mu_t = \mu_{t-1} + \eta_t, \quad \eta_t \sim \mathcal{N}(0, q),$$

and observe returns through

$$r_t = \mu_t + \xi_t, \quad \xi_t \sim \mathcal{N}(0, \sigma^2).$$

Here $q \geq 0$ is the drift-adaptation variance. A larger q makes the drift estimate react faster to recent returns, while $q = 0$ gives a constant-drift estimator whose gain shrinks over time.

To avoid lookahead bias, the funnel for step s uses the one-step-ahead prediction $\hat{\mu}_{s|s-1}$, not the posterior estimate after seeing r_s . The Kalman recursion is

$$\hat{\mu}_{s|s-1} = \hat{\mu}_{s-1|s-1}, \quad P_{s|s-1} = P_{s-1|s-1} + q,$$

$$K_s = \frac{P_{s|s-1}}{P_{s|s-1} + \sigma^2},$$

$$\hat{\mu}_{s|s} = \hat{\mu}_{s|s-1} + K_s (r_s - \hat{\mu}_{s|s-1}), \quad P_{s|s} = (1 - K_s)P_{s|s-1}.$$

With adaptive drift and fixed volatility, the Phase 4 funnel after reference time τ becomes

$$C_\tau(t) = X_\tau + \sum_{s=\tau+1}^t \hat{\mu}_{s|s-1}, \quad V_\tau(t) = \sum_{s=\tau+1}^t \sigma^2.$$

The simulation initializes the filter from a neutral prior rather than the true simulated drift. The same upper-funnel sell rule, lower-funnel re-entry rule, momentum filter, and transaction-cost profit threshold are then applied using this adaptive center.

15 Phase 5 Adaptive Volatility Estimation

Phase 5 removes the fixed-width volatility assumption from the funnel. Instead of using $V_\tau(t) = \sigma^2(t - \tau)$, the strategy estimates a one-step-ahead conditional variance and accumulates those forecasts over the life of each local funnel.

Let the innovation relative to the drift forecast be

$$e_s = r_s - \hat{\mu}_{s|s-1}.$$

A GARCH(1,1)-style conditional variance recursion is

$$\hat{\sigma}_{s|s-1}^2 = h_s,$$

$$h_{s+1} = \omega + \alpha e_s^2 + \beta h_s, \quad \alpha \geq 0, \quad \beta \geq 0, \quad \alpha + \beta < 1.$$

The parameter α controls how strongly the variance forecast reacts to the most recent shock, while β controls volatility persistence. In the simulation, ω is chosen from an initial long-run variance level:

$$\omega = (1 - \alpha - \beta)\bar{\sigma}^2.$$

To avoid lookahead bias, the funnel for step s uses $\hat{\sigma}_{s|s-1}^2$, the variance forecast available before observing r_s . With adaptive drift and adaptive volatility, the Phase 5 funnel after reference time τ is

$$C_\tau(t) = X_\tau + \sum_{s=\tau+1}^t \hat{\mu}_{s|s-1}, \quad V_\tau(t) = \sum_{s=\tau+1}^t \hat{\sigma}_{s|s-1}^2.$$

The standardized deviation is therefore

$$Z_t = \frac{X_t - C_\tau(t)}{\sqrt{V_\tau(t)}}.$$

This form is important: since $C_\tau(t)$ already contains the anchor X_τ , subtracting both X_τ and $C_\tau(t)$ would double-count the reference price.

Adaptive volatility makes the funnel breathe. After large shocks, $V_\tau(t)$ grows faster and the upper and lower boundaries widen; during quieter periods, the funnel narrows. This usually reduces false boundary crossings in turbulent regimes and makes the visual funnel closer to how market uncertainty changes over time.

16 Phase 6 Walk-Forward Backtesting

Phase 6 changes the evaluation question from “does the rule behave correctly on a controlled path?” to “does the adaptive rule survive out-of-sample testing?” The essential requirement is that parameters are chosen using only past data and then evaluated on future data that was not used for selection.

Let the observed price history be divided into ordered train/test folds. For fold j , write

$$\mathcal{T}_j = \{t_j, \dots, t_j + L_{\text{train}}\}$$

for the training window and

$$\mathcal{O}_j = \{t_j + L_{\text{train}}, \dots, t_j + L_{\text{train}} + L_{\text{test}}\}$$

for the following out-of-sample window. Candidate parameters such as k , δ , Kalman process variance q , and GARCH parameters may be compared on $\mathcal{T}_j$, but once the best candidate is selected, it is frozen before running the strategy on $\mathcal{O}_j$.

For a candidate parameter vector θ , define a training objective such as

$$J_j(\theta) = R_{j,\text{train}}^m(\theta) + D_{j,\text{train}}^m(\theta) + \lambda G_{j,\text{train}}(\theta),$$

where R^m is mark-to-market return, $D^m \leq 0$ is maximum drawdown, G is the benchmark gap, and $\lambda \geq 0$ controls how strongly benchmark-relative performance is rewarded. The selected parameter is

$$\hat{\theta}_j = \arg \max_{\theta \in \Theta} J_j(\theta).$$

The out-of-sample fold result is then computed by running the strategy on $\mathcal{O}_j$ using $\hat{\theta}_j$, with the adaptive drift and volatility filters initialized only from information available at the beginning of that test fold.

The out-of-sample fold multipliers can be compounded:

$$M_{\text{OOS}} = \prod_j \frac{W_{j,\text{end}}^m}{W_{j,\text{start}}^m}, \quad R_{\text{OOS}} = M_{\text{OOS}} - 1.$$

Additional diagnostics should include maximum drawdown, exposure, completed trade count, benchmark gap, Sharpe ratio, and Calmar ratio. Using wealth values W_t , the annualized Sharpe ratio with zero risk-free rate is

$$\text{Sharpe} = \sqrt{252} \frac{\bar{r}}{s_r}, \quad r_t = \frac{W_t - W_{t-1}}{W_{t-1}},$$

and the Calmar ratio is

$$\text{Calmar} = \frac{R_{\text{ann}}}{|D_{\text{min}}|}.$$

This protocol does not prove profitability. Its purpose is more modest and more important: it prevents accidental lookahead bias, separates parameter fitting from evaluation, and gives a fairer comparison against buy-and-hold. In the implementation, a local regime-changing path is used as an offline walk-forward demonstration. The same logic is also applied to real daily market prices fetched from a public chart API or loaded from an adjusted-close CSV file. The data source, symbol, sampling frequency, and whether prices are adjusted for dividends and splits should be reported with every real-data result.

For the real daily-price visualization, the implementation uses a continuous online variant rather than restarting the strategy at every fold. The strategy begins with initial parameters $\theta_0 = (k_0, \delta_0)$. At the start of each new trading week, the most recent week of daily prices is used to compute a suggested parameter vector $\tilde{\theta}_j$. The live parameter is then updated by smoothing:

$$\theta_j = (1 - \rho)\theta_{j-1} + \rho\tilde{\theta}_j, \quad 0 < \rho \leq 1.$$

For integer-valued parameters such as δ , the smoothed value is rounded and constrained to the candidate grid. The next week is traded using θ_j , while the position state, realized wealth, and mark-to-market wealth continue without reset. This gives a more realistic animation: one continuous trading account learns from recent data while avoiding lookahead.

17 Discussion and Limitations

The strategy enforces a no-realized-loss condition on completed voluntary exits. However, this does not imply that the strategy is risk-free. The main limitations are:

- **Mark-to-market drawdown:** the current value of the position may fall while holding;
- **Indefinite holding:** if the stopping rule never triggers, the position may remain open indefinitely;
- **Forced-exit risk:** finite-horizon liquidation can realize losses;
- **Opportunity cost:** the strategy may underperform buy-and-hold;
- **Model misspecification:** real prices have changing volatility, jumps, and fat tails;
- **Parameter estimation error:** the drift μ is difficult to estimate accurately.

Future phases should replace the offline demonstration path with cleaned real adjusted-close data and report the full walk-forward fold table.

18 Conclusion

We formulated a funnel-based stopping strategy for trading under drift and volatility. The key correction is the explicit profit threshold h , which gives a rigorous no-realized-loss theorem for completed algorithmic exits. The strategy remains risky in mark-to-market terms, and its practical value must be judged by simulation, buy-and-hold comparison, drawdown, and holding-time behavior.